Library Management System

Assignment 1 – Python Program with Flowchart and Output

Roll Number: 34

Enrollement no.:ADT25SOCB0817

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1. Python Code

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"""
LIBRARY MANAGEMENT SYSTEM
Implements the flowchart exactly:

START
 -> Create Library Object
 -> LOOP: User Selects Operation
      1. Add Book        -> Add Book to Library
      2. Register Patron  -> Add Patron to Library
      3. View Books & Patrons -> Display Books and Patrons
      4. Borrow Book -> Book Available?
            Yes -> Issue Book to Patron / Update Status
            No  -> Display "Not Available"
      5. Return Book -> Book Borrowed by Patron?
            Yes -> Accept Return / Update Status
            No  -> Display "Book not borrowed by this patron"
 -> Continue?
      Yes -> loop back to "User Selects Operation"
      No  -> END
"""


class Book:
    def __init__(self, isbn, title, author):
        self.isbn = isbn
        self.title = title
        self.author = author
        self.available = True          # Book Available? state
        self.borrowed_by = None        # tracks which patron holds it

    def __str__(self):
        status = "Available" if self.available else f"Borrowed by {self.borrowed_by}"
        return f"[{self.isbn}] '{self.title}' by {self.author} - {status}"


class Patron:
    def __init__(self, patron_id, name):
        self.patron_id = patron_id
        self.name = name

    def __str__(self):
        return f"[{self.patron_id}] {self.name}"


class Library:
    """Create Library Object"""

    def __init__(self):
        self.books = {}     # isbn -> Book
        self.patrons = {}   # patron_id -> Patron

    # 1. Add Book -> Add Book to Library
    def add_book(self, isbn, title, author):
        self.books[isbn] = Book(isbn, title, author)
        print(f"Book added: {self.books[isbn]}")
    # 2. Register Patron -> Add Patron to Library
```

```
    def register_patron(self, patron_id, name):
        self.patrons[patron_id] = Patron(patron_id, name)
        print(f"Patron registered: {self.patrons[patron_id]}")

    # 3. View Books & Patrons -> Display Books and Patrons
    def view_books_and_patrons(self):
        print("\n--- Books ---")
        if not self.books:
            print("No books in library.")
        for book in self.books.values():
            print(book)

        print("\n--- Patrons ---")
        if not self.patrons:
            print("No patrons registered.")
        for patron in self.patrons.values():
            print(patron)

    # 4. Borrow Book -> Book Available? -> Yes/No branches
    def borrow_book(self, isbn, patron_id):
        book = self.books.get(isbn)

        if book is None:
            print("Display: 'Not Available' (no such book)")
            return

        if patron_id not in self.patrons:
            print("Display: 'Not Available' (unknown patron)")
            return

        if book.available:                      # Book Available? -> Yes
            book.available = False               # Issue Book to Patron / Update Status
            book.borrowed_by = patron_id
            print(f"Issued '{book.title}' to patron {patron_id}. Status updated.")
        else:                                    # Book Available? -> No
            print("Display: 'Not Available'")

    # 5. Return Book -> Book Borrowed by Patron? -> Yes/No branches
    def return_book(self, isbn, patron_id):
        book = self.books.get(isbn)

        if book is None:
            print("Display: 'Book not borrowed by this patron' (no such book)")
            return

        if not book.available and book.borrowed_by == patron_id:   # Yes
            book.available = True                                   # Accept Return / Update Status
            book.borrowed_by = None
            print(f"Return accepted for '{book.title}'. Status updated.")
        else:                                                        # No
            print("Display: 'Book not borrowed by this patron'")


def user_selects_operation():
    print("\n===== USER SELECTS OPERATION =====")
    print("1. Add Book")
    print("2. Register Patron")
    print("3. View Books & Patrons")
    print("4. Borrow Book")
    print("5. Return Book")
    return input("Choose an option (1-5): ").strip()

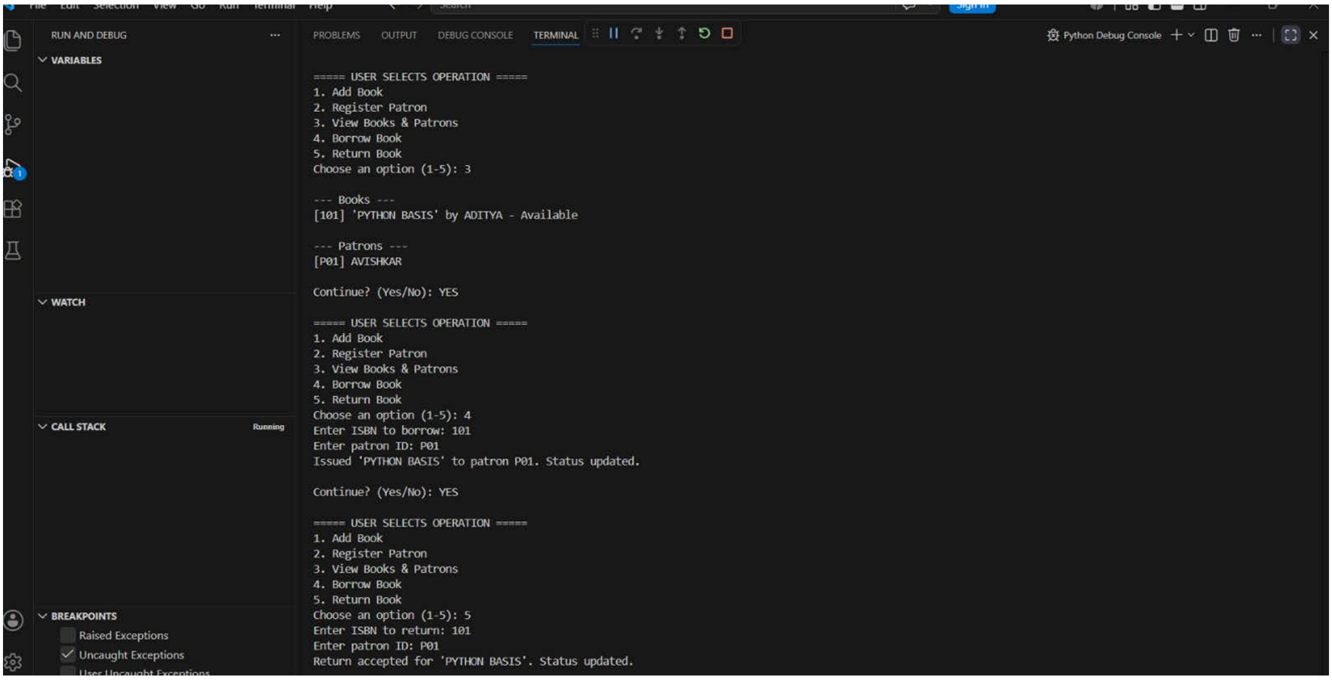

def main():
    # START -> Create Library Object
    library = Library()

    while True:  # loop back point for "Continue? Yes"
        choice = user_selects_operation()
```

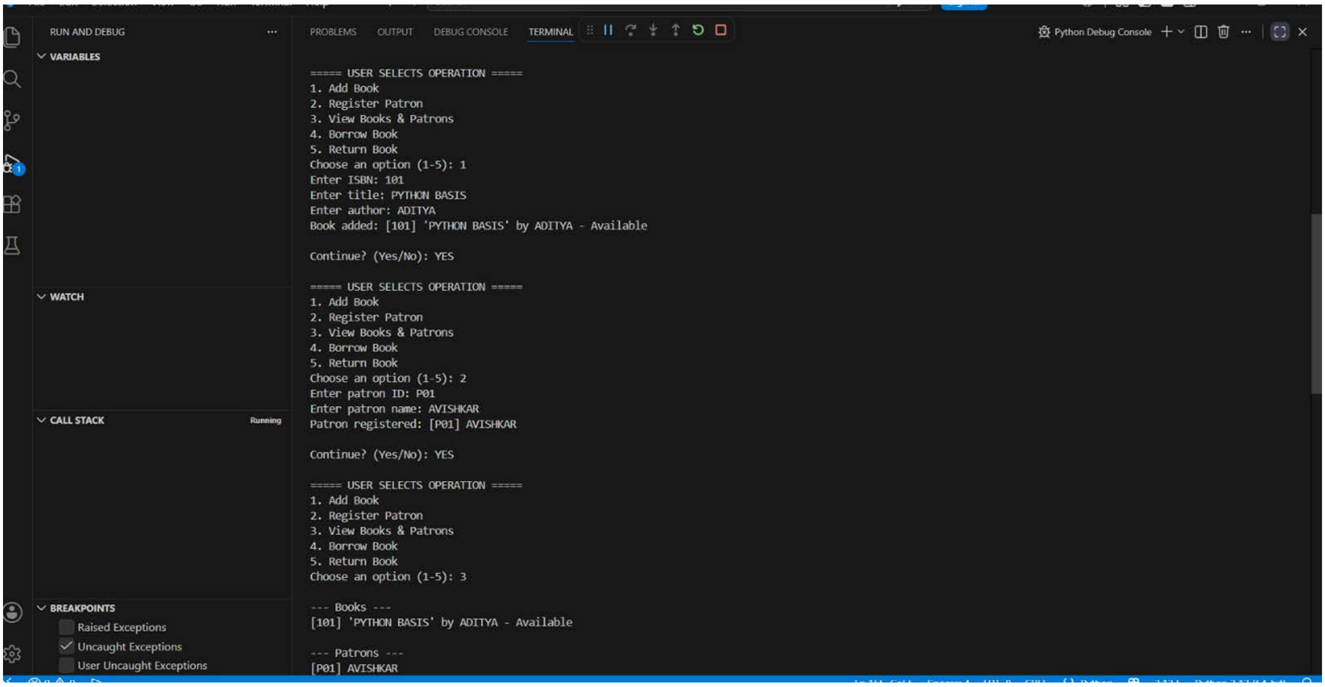
```python
        if choice == "1":
            isbn = input("Enter ISBN: ").strip()
            title = input("Enter title: ").strip()
            author = input("Enter author: ").strip()
            library.add_book(isbn, title, author)

        elif choice == "2":
            patron_id = input("Enter patron ID: ").strip()
            name = input("Enter patron name: ").strip()
            library.register_patron(patron_id, name)

        elif choice == "3":
            library.view_books_and_patrons()

        elif choice == "4":
            isbn = input("Enter ISBN to borrow: ").strip()
            patron_id = input("Enter patron ID: ").strip()
            library.borrow_book(isbn, patron_id)

        elif choice == "5":
            isbn = input("Enter ISBN to return: ").strip()
            patron_id = input("Enter patron ID: ").strip()
            library.return_book(isbn, patron_id)

        else:
            print("Invalid option. Please choose 1-5.")

        # Continue? decision
        cont = input("\nContinue? (Yes/No): ").strip().lower()
        if cont in ("no", "n"):
            break  # -> END

    print("\nEND")


if __name__ == "__main__":
    main()
```

2. Program Output

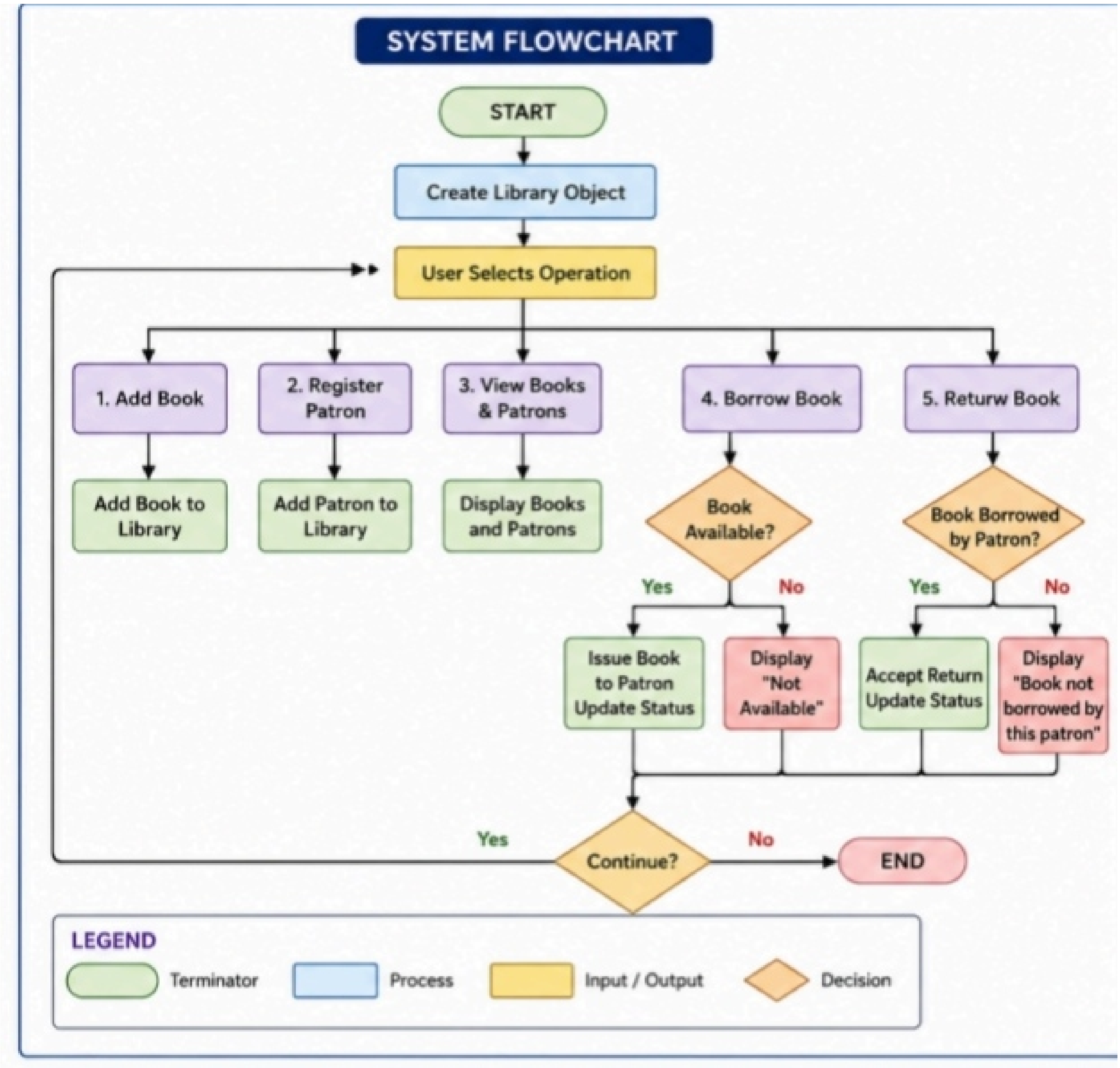
Output Screenshot 1 - Add Book, Register Patron, View Books & Patrons



Output Screenshot 2 - Borrow Book and Return Book



3. System Flowchart



4. Algorithm

Step 1: Start the program.

Step 2: Create a Library object to store books and patrons.

Step 3: Repeat Steps 4 to 10 until the user chooses to stop.

Step 4: Display the menu and let the user select an operation (1–5).

Step 5: If choice = 1 (Add Book) → read ISBN, title and author, then add the book to the library.

Step 6: If choice = 2 (Register Patron) → read patron ID and name, then add the patron to the library.

Step 7: If choice = 3 (View Books & Patrons) → display the list of all books and all patrons.

Step 8: If choice = 4 (Borrow Book) → read ISBN and patron ID, then check if the book is available. If available, issue the book to the patron and update its status. If not available, display "Not Available".

Step 9: If choice = 5 (Return Book) → read ISBN and patron ID, then check if the book was borrowed by that patron. If yes, accept the return and update its status. If no, display "Book not borrowed by this patron".

Step 10: Ask the user "Continue?" (Yes/No). If Yes, go back to Step 4. If No, go to Step 11.

Step 11: Stop the program (End).